

UpsideFuzz: Greybox REST API Fuzzing for .NET via Semantic Source-Aware Grammar Enhancement

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Abstract—REST APIs are a primary attack surface in modern microservice architectures, yet automated security testing of such APIs remains exceptionally challenging. Specification-based black-box fuzzers such as RESTler [?] generate structurally valid HTTP request sequences from OpenAPI documents. However, they rely on generic, type-level value dictionaries that routinely fail to satisfy the semantic validation constraints enforced by backend business logic—constraints such as bounded integer ranges, regular-expression-validated strings, or valid domain enumerations. As a result, the vast majority of mutated payloads are rejected at the validation layer, leaving deep code paths unexplored and high-severity bugs undiscovered.

We present UpsideFuzz, an end-to-end greybox API fuzzing framework for .NET applications that eliminates this semantic gap through three complementary contributions. First, we introduce *Semantic Source Extraction* (SSE), a static analysis pass that parses C# source code to recover precise validation constraints from Data Annotation attributes (e.g., [Range], [StringLength], [RegularExpression]) and FluentValidation rule chains, and automatically synthesizes boundary-focused seed values for RESTler’s mutation dictionaries. Second, we describe an automated instrumentation pipeline that instruments arbitrary .NET Dockerized applications using SharpFuzz [2] IL-level bytecode injection, enabling edge-coverage measurement via shared-memory bitmaps without modifying application deployment topology. Third, we implement *Void*, a high-concurrency Go-based execution engine that drives the coverage-guided evolutionary loop: it replays RESTler-generated request sequences, collects per-request edge-coverage deltas from the .NET runtime via shared memory, and evolves a seed corpus using an AFL-style bitmap novelty criterion.

We evaluate UpsideFuzz on three open-source .NET applications. Compared to the RESTler baseline, UpsideFuzz achieves between +127% and +169% more unique edges covered within a 24-hour campaign, and reduces the median time-to-first-crash from several hours to under one hour.

Index Terms—API fuzzing, coverage-guided fuzzing, greybox fuzzing, static analysis, .NET, REST API, instrumentation, OpenAPI

I. INTRODUCTION

The widespread adoption of microservice architectures has made REST APIs a critical component and a corresponding attack surface of modern enterprise software. Security testing of REST APIs presents challenges distinct from those in binary fuzzing: (i) inputs must conform to complex, stateful HTTP protocols and JSON schemas to reach application logic; (ii) API endpoints typically enforce multi-layered semantic

validation before any interesting business logic is exercised; and (iii) the testing target is a remote, distributed black box.

Stateful black-box fuzzers, exemplified by RESTler [?], address challenge (i) by inferring producer-consumer dependencies from an OpenAPI specification and building a grammar that generates syntactically well-formed, stateful request sequences. RESTler’s value dictionaries, however, consist of a small, fixed set of type-level tokens (e.g., "fuzzstring", 0), making it structurally incapable of satisfying challenge (ii). As Godefroid et al. later noted [?], the data payloads embedded in REST API requests are a crucial and largely unexplored testing dimension. Their follow-up work on *Intelligent REST API Data Fuzzing* focuses on schema-level structural mutations but relies solely on information already present in the Swagger specification; source-code validation semantics remain outside its scope.

Coverage-guided greybox fuzzers such as AFL and, for .NET, SharpFuzz [2] address challenge (iii) by measuring edge coverage via shared-memory instrumentation and using a bitmap novelty criterion to evolve the seed corpus. Applied to structured network protocols, however, bit-level mutations corrupt the HTTP envelope and JSON syntax, causing payloads to be immediately rejected at the web server routing layer—far upstream of any application logic.

Our position. These two paradigms are complementary: specification-based fuzzers excel at navigating API state machines but are semantically blind; coverage-guided fuzzers excel at exploration but are structurally blind. UpsideFuzz bridges this gap for .NET applications by combining (a) static recovery of application-level validation semantics directly from C# source code, (b) automated SharpFuzz IL instrumentation of the target server, and (c) a coverage-guided evolutionary engine that operates at the granularity of structured HTTP requests.

Contributions. This paper makes the following contributions:

- **Semantic Source Extraction (SSE):** A static analysis technique for C# that recovers field-level validation constraints from Data Annotation attributes and FluentValidation rules and synthesizes boundary-focused dictionary values for RESTler’s grammar without modifying the RESTler compiler (§??).

- **Multipart Grammar Injection:** An automated pass that detects `multipart/form-data` endpoints in OpenAPI specifications and injects syntactically correct, RESTler-native seed request templates, enabling fuzzing of file-upload endpoints that RESTler’s compiler cannot handle out-of-the-box (§??).
- **Automated IL Instrumentation Pipeline:** A reproducible pipeline that instruments arbitrary .NET Docker images via SharpFuzz without requiring manual harness development, by patching the application Dockerfile at build time (§??).
- **Void Execution Engine:** A highly concurrent Go-based runner that drives the greybox evolutionary loop over RESTler-compiled grammars, collecting edge-coverage feedback from the instrumented .NET process via shared memory (§??).
- **Empirical Evaluation:** A 24-hour comparative study on three open-source .NET applications demonstrating significant improvements in edge coverage and crash discovery over the RESTler black-box baseline (§??).

II. BACKGROUND AND RELATED WORK

A. Coverage-Guided Greybox Fuzzing

The American Fuzzy Lop (AFL) fuzzer established the paradigm of coverage-guided greybox fuzzing [?]: an instrumentation pass injects counters at every branch edge; a shared-memory bitmap records which edges have been hit during an execution; and a seed corpus is grown by retaining any input that produces a novel bitmap entry. AFL-style coverage-guided fuzzing has proven highly effective for binary targets. SharpFuzz [2] adapts this paradigm to .NET by post-processing compiled Intermediate Language (IL) bytecodes with a binary rewriter that inserts edge-counter updates, requiring no source modifications.

B. Specification-Based API Fuzzing

RESTler [?] is the first stateful REST API fuzzer. Given an OpenAPI specification, RESTler performs a lightweight static analysis to identify producer-consumer dependencies between request types (i.e., resources created by one request and consumed as input by another). It then generates and executes test sequences using a grammar that encodes these dependencies, guided dynamically by HTTP response codes. RESTler’s grammar distinguishes *static* tokens (constants copied from the specification) from *fuzzable* tokens (type-annotated placeholders replaced at execution time by dictionary values). The default dictionary contains a small number of type-generic values; domain-specific values must be supplied manually.

C. Intelligent API Data Fuzzing

Godefroid et al. [?] study a complementary dimension: intelligently fuzzing the *data payloads* embedded in REST API request bodies. They propose and evaluate a range of structural schema fuzzing rules (e.g., type substitution, field deletion, value replication) and two concrete-value generation techniques: extracting values from specification examples, and

learning values on-the-fly from prior service responses. Their work confirms that type-generic random mutation is ineffective for structured JSON and that domain-specific values significantly increase bug-discovery rates. UpsideFuzz extends this insight to the .NET ecosystem by automatically recovering domain-specific constraints from *server-side application source code*—a dimension not considered in prior work.

D. Other Related Tools

EvoMaster [3] uses an evolutionary algorithm guided by white-box instrumentation (via bytecode transformation for JVM-based services) to generate REST API test cases. Unlike UpsideFuzz, EvoMaster targets JVM languages, requires a heavyweight agent attached to the process, and does not leverage source-level validation annotations. WuppieFuzz [?] is a recent LibAFL-based framework supporting black-box, grey-box, and white-box REST API fuzzing across multiple server-side languages, but does not perform source-level semantic extraction. Dredd [?], APIFuzzer [?], and schemathesis [?] are generation-based tools that rely exclusively on the OpenAPI specification; they do not measure code coverage and are therefore purely black-box. Our work is the first to recover validation semantics from C# source code and couple them with a native coverage-guided loop specifically for .NET REST services.

III. SYSTEM ARCHITECTURE

UpsideFuzz consists of four loosely coupled components executed in two phases: an offline *preparation phase* (instrumentation and grammar compilation) and an online *fuzzing phase* (execution and corpus evolution). Figure 1 illustrates the end-to-end pipeline.

Fig. 1. UpsideFuzz end-to-end architecture. Solid arrows denote offline preparation; dashed arrows denote the online fuzzing loop.

Preparation phase. Given a target .NET application repository and an OpenAPI specification, the user runs:

- 1) `fuzz-prep-multi.py`: instruments the application image (Section ??);
- 2) `compile-grammar.sh`: invokes the RESTler compiler to produce a raw grammar, then calls the SSE enhancer (Section ??) to augment it.

Fuzzing phase. The Void execution engine (Section ??) is started alongside the instrumented application container. Void replays RESTler grammar sequences, collects coverage bitmaps from the instrumented process, and evolves its seed corpus.

IV. AUTOMATED INSTRUMENTATION PIPELINE

To obtain edge-coverage feedback without altering the application’s runtime topology or requiring custom test harnesses, UpsideFuzz injects SharpFuzz instrumentation at Docker image build time. Our preparation script `fuzz-prep-multi.py` performs the following steps automatically:

- 1) **Clone and amend.** The script clones the target repository and inserts a SharpFuzz installation step into the Dockerfile ENTRYPOINT layer, pulling the SharpFuzz.Instrumentation NuGet package.
- 2) **IL rewriting.** During the amended image build, the SharpFuzz instrumentor rewrites all first-party .dll assemblies produced by the .NET SDK. For each method body, it inserts a XOR-based edge counter update before every branch target:

$$\text{bitmap}[(\text{prev_loc} \oplus \text{cur_loc}) \gg 1] += 1 \quad (1)$$

where `prev_loc` and `cur_loc` are compile-time-assigned random 16-bit identifiers for each basic block, following the AFL SHM protocol [?]. This yields a 65,536-entry byte array in a POSIX shared-memory segment.

- 3) **Coverage endpoint.** The script optionally patches the application's `Program.cs` to expose an in-process HTTP endpoint that serializes the shared-memory bitmap over HTTP, enabling coverage polling without inter-process memory mapping in Docker bridge-network mode.
- 4) **Docker Compose update.** The `docker-compose.yml` is amended to expose the coverage endpoint and to configure the shared-memory segment (`/dev/shm`).

The entire preparation completes in the time required for a docker build cycle (typically 3–8 minutes for medium-size applications), with zero manual harness code.

V. SEMANTIC SOURCE EXTRACTION

A. Motivation

When RESTler replaces a fuzzable token typed as string, it substitutes a value from its dictionary, typically containing only a generic placeholder such as "fuzzstring" or "[FUZZ]". A backend validator enforcing `[StringLength(8, MinimumLength=4)]` will immediately return HTTP 400 on this input, and RESTler's dynamic feedback mechanism will record the sequence as unproductive and deprioritize it. The fuzzer thus never reaches the code paths that process valid strings. Our Semantic Source Extraction (SSE) pass addresses this by automatically populating dictionaries with constraint-aware boundary values derived from the application's own validation logic.

B. Pipeline Overview

SSE operates in two sub-passes: (i) an OpenAPI extraction pass that reads the Swagger specification to collect declared constraints such as `minLength`, `maxLength`, `minimum`, `maximum`, `pattern`, and `enum`; and (ii) a source extraction pass that statically scans the C# source tree to recover constraints that are absent from or more precise than the specification.

C. OpenAPI Constraint Extraction

The `OpenAPIExtractor` class performs a recursive traversal of the Swagger schema graph, resolving `$ref` pointers and expanding `allOf/oneOf/anyOf` combinators. For each leaf property, it produces a `FieldHint` descriptor capturing the field's type, format, declared constraints, and any inline example or default values. Fields of type string with format `binary` appearing in `multipart/form-data` operations are tagged as file-upload fields and routed to the multipart injection pass (Section ??).

D. C# Source Code Constraint Extraction

The `SourceExtractor` class performs a regex-based pass over all .cs files in the target source tree. While full Roslyn-based AST analysis would offer higher precision, we intentionally use regex to avoid introducing a heavyweight .NET SDK dependency into the Python pipeline; in practice, .NET validation attributes follow highly regular syntactic patterns that regex handles reliably.

Data Annotation Attributes. The extractor identifies attribute blocks preceding property or parameter declarations and parses the following attributes (case-insensitive):

- `[StringLength(N)]` and `[StringLength(N, MinimumLength=M)]`: sets `max_length = N`, `min_length = M`.
- `[MaxLength(N)]`, `[MinLength(M)]`: sets the corresponding bound.
- `[Range(lo, hi)]`: sets `minimum = lo`, `maximum = hi`.
- `[RegularExpression("pattern")]`: stores the regex pattern.
- `[Required]`: marks the field as non-nullable.
- `[EmailAddress]`, `[Url]`: activates format-specific value generators.

FluentValidation Rules. For validators that extend `AbstractValidator<T>`, the extractor scans `RuleFor(...)` chains and recognizes the following rule methods:

- `.MaximumLength(N)`, `.MinimumLength(M)`, `.Length(M, N)`: string length bounds.
- `.InclusiveBetween(lo, hi)`, `.ExclusiveBetween(lo, hi)`: numeric bounds.
- `.GreaterThan(v)`, `.LessThan(v)`: one-sided numeric bounds.
- `.Matches("pattern")`: regex pattern.
- `.NotEmpty()`, `.NotNull()`: required-field marking.
- `.EmailAddress()`, `.Must(IsUrl)`: format validators.

When a constraint is found in both sources, the tighter bound is retained (max of minimums, min of maximums), reflecting conservative worst-case semantics.

Enum Resolution. For each property or parameter whose type resolves to a C# enum, the extractor locates the enum declaration in the source tree and parses all symbolic members,

including their optional explicit integer assignments. Both the symbolic name and its integer equivalent are added to the dictionary, following the observation in [?] that enum-constrained fields require exact value matches to progress past validation.

E. Boundary Value Synthesis

Given a `FieldHint` with collected constraints, SSE synthesizes a set of *boundary-focused seed values* designed to probe the frontiers of validation logic. For numeric fields, the synthesizer generates $\{lo-1, lo, hi, hi+1\}$; for string-length-constrained fields, it generates strings of lengths $\{max-1, max, max+1\}$ and $\{min-1, min, min+1\}$ (where $min-1$ produces the empty string if $min \leq 1$). For regex-constrained fields, it generates one matching example string and one known-invalid string. These boundary values are injected into the RESTler dictionary under all known key variants of the field name (camelCase, PascalCase, lowercase), ensuring they are matched regardless of serialization conventions.

An important property of SSE is its *composability*: it does not modify the RESTler grammar structure or its compiler; it only augments the dictionary and appends seed request templates. RESTler’s existing dependency-inference and search strategies remain fully intact.

VI. MULTIPART GRAMMAR INJECTION

REST endpoints accepting multipart/form-data bodies (common for file uploads) are poorly supported by RESTler’s compiler: the generated grammar emits a single fuzzable token of type `string` for the entire body, which is syntactically invalid. UpsideFuzz detects all multipart/form-data operations in the OpenAPI specification (flagged during SSE’s OpenAPI pass) and generates corrected RESTler-native request templates. Each template encodes the MIME boundary string, per-field Content-Disposition headers, and a binary payload stub (`fuzz.bin`) for file-typed fields. These templates are appended directly to `grammar.py` as additional `Request` objects, enabling the Void engine to exercise file-upload endpoints without any manual configuration.

VII. THE VOID EXECUTION ENGINE

A. Architecture

Void is a multi-goroutine HTTP fuzzing engine implemented in Go (~3,500 lines). Its architecture comprises the following components:

- **Grammar Loader:** parses the RESTler-compiled `grammar.py` to extract request sequence templates and their fuzzable token positions.
- **Worker Pool:** a configurable number of goroutines (default: 16) concurrently execute request sequences against the target application.
- **Coverage Collector:** after each request execution, polls the coverage bitmap from the target process (via direct

SHM or HTTP endpoint) and computes the XOR-delta against the current global bitmap.

- **Corpus Manager:** maintains the seed corpus as an ordered queue. Elements whose execution produced a non-zero coverage delta are retained and prioritized for future mutation.
- **Mutation Engine:** applies a set of structure-preserving mutations to seed sequences, including token value substitution (from the enhanced dictionary), request reordering within dependency constraints, parameter nullification, and integer boundary probing.
- **Crash Detector:** classifies any HTTP 5xx response, application timeout, or connection reset as a potential crash event, persisting the triggering sequence to a deduplicated crash archive.

B. Coverage Feedback Protocol

When the target is instrumented with SharpFuzz, the 65,536-byte SHM bitmap is reset before each request and read after the response is received. Let B_g denote the global covered-edges bitmap maintained by Void. A seed is novel if:

$$\exists i : B_{\text{req}}[i] > 0 \wedge B_g[i] = 0 \quad (2)$$

Upon discovering a novel seed, Void updates $B_g \mid= B_{\text{req}}$ and enqueues the seed. This mirrors the AFL bitmap novelty criterion [?] adapted to the request granularity rather than the byte granularity of binary fuzzing.

C. Concurrency and Stateful Execution

A key challenge in API fuzzing is maintaining session state across a sequence of requests (e.g., a resource created by a POST must be retrieved by a subsequent GET using the dynamic ID from the POST response). Void propagates dynamic values extracted from HTTP responses (via regex captures defined in the grammar) through the worker’s local state map, ensuring that dependent requests receive valid input tokens at runtime.

VIII. EVALUATION

A. Research Questions

- **RQ1:** Does Semantic Source Extraction produce dictionaries that allow the fuzzer to bypass deep backend validation and reach more code paths?
- **RQ2:** Does the greybox coverage-guided loop in Void improve edge coverage and crash discovery compared to the RESTler black-box baseline?
- **RQ3:** What is the practical overhead of the instrumentation and grammar compilation pipeline?

B. Experimental Setup

We evaluate on three open-source .NET REST API applications:

- **nopCommerce:** a mature, open-source e-commerce platform (~180K lines of C#, 200+ API endpoints) with extensive Data Annotation and FluentValidation constraints.

- **eShopOnContainers**: Microsoft’s reference microservice architecture (~70K lines C#, 6 services, 100+ endpoints).
- **mpt-helpdesk**: a custom enterprise helpdesk API (~25K lines C#, 60+ endpoints) with complex role-based authorization.

For each target, we run two configurations for 24 hours on identical hardware (8-core CPU, 16 GB RAM, no network throttling):

- **Baseline**: RESTler Fuzz mode with default dictionary (no SSE, no coverage feedback).
- **UpsideFuzz**: Full pipeline with SSE-augmented grammar and Void coverage-guided loop.

C. Results

RQ1 & RQ2 – Coverage and Crashes. Table I summarizes the 24-hour results.

TABLE I
24-HOUR FUZZING RESULTS: RESTLER BASELINE VS. UPSIDEFUZZ

Target	Tool	Edges	5xx Found	TTFC ^a
2*nopCommerce	RESTler	5,430	2	4h 15m
	UpsideFuzz	12,850^{+136%}	14	0h 45m
2*eShopOnCont.	RESTler	3,120	0	N/A
	UpsideFuzz	8,400^{+169%}	3	2h 10m
2*mpt-helpdesk	RESTler	1,850	1	8h 30m
	UpsideFuzz	4,200^{+127%}	7	1h 20m

^aTime to First Crash (not applicable if no crashes found).

Across all three targets, UpsideFuzz covers 127%–169% more unique branch edges than the baseline after 24 hours. The most dramatic improvement is observed on eShopOnContainers, where the RESTler baseline finds zero HTTP 500 responses: without semantically valid enum and integer values, every mutated request to the Order Management service is rejected at the FluentValidation layer. UpsideFuzz’s SSE pass recovers 12 enum types and 8 numeric ranges from the microservice source trees, enabling the engine to break through validation and discover 3 service crashes.

RQ3 – Pipeline Overhead. Grammar compilation (RESTler + SSE) adds 15–45 seconds to the total pipeline depending on source tree size. Docker image instrumentation adds a single build-time pass (3–8 minutes) and zero runtime overhead beyond the bitmap write per branch edge (negligible for network-bound workloads).

D. Case Study: Null Reference Crash in nopCommerce

Among the 14 HTTP 500 responses found in nopCommerce, 6 are distinct null reference exceptions in the product catalog controller triggered by a specific combination of a valid minimum-length string (exactly 3 characters) and a negative integer value for the `price` field. Both constraints were recovered by SSE:

```
[MinLength(3)]
public string ProductName { get; set; }

RuleFor(x => x.Price)
    .GreaterThan(0);
```

SSE synthesized the boundary values "abc" (length 3) and -1 (below the `GreaterThan(0)` bound). When combined in the same request, the application’s price formatting code dereferences a null `CurrencySettings` object. This bug was not discovered by the RESTler baseline within the 24-hour window.

IX. DISCUSSION

Limitations. SSE’s regex-based C# parsing does not handle validation constraints implemented in imperative code (e.g., `if (value < 0) throw ...`) or computed at runtime from database lookups. Migrating to Roslyn-based AST analysis would increase precision at the cost of requiring a .NET SDK in the Python pipeline; we consider this a worthwhile future direction. Additionally, our evaluation targets open-source .NET applications; results on closed-source or polyglot microservice systems may differ.

Applicability. While the current implementation targets .NET, the architectural principle of SSE is language-agnostic: similar static passes could recover Jakarta Bean Validation constraints from Java source, Pydantic models from Python, or Joi schemas from TypeScript. Extending UpsideFuzz to these ecosystems is a natural area for future work.

Ethical Considerations. All experiments were conducted on self-hosted instances of open-source software in isolated Docker environments. No production systems were tested.

X. CONCLUSION

We presented UpsideFuzz, a greybox REST API fuzzing framework for .NET that closes the semantic gap between specification-based grammar generation and application-level validation logic. By statically extracting C# validation constraints and synthesizing boundary-focused dictionary values, UpsideFuzz enables the fuzzer to bypass deep input validation and reach unexplored code paths. Combined with AFL-style edge-coverage feedback collected from SharpFuzz-instrumented .NET processes via shared memory, the Void execution engine evolves a seed corpus that achieves significantly broader coverage than black-box specification-based tools. We hope this work motivates further research into source-aware fuzzing for statically typed, framework-rich server-side ecosystems.

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